Learned representations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable representation and collision record.

Trace preserved/lost information,repair available measurements and state the learning-evidence limit.

Keep one habit: W1–W3/W5 → selected new encoder → project representation ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Consider fictional rectangles with **input** [width, height] in cm. The task is to identify rectangles whose width is strictly greater than height. The target is 1 for wider rectangles and 0 otherwise.

  A square therefore has target 0. This is an arithmetic teaching task, not a claim that a network is needed to measure rectangles.

- Define two hidden units: h1 = ReLU(width - height) and h2 = ReLU(height - width). Their weight rows are [1, -1] and [-1, 1], with zero biases. The **representation** h = [h1, h2] records excess width or excess height.

  Predict 1 when h1 > 0; otherwise predict 0.

- These weights were chosen by the lesson author. They demonstrate a **representation** that could be implemented by a network, but no **training** run has established that a model learned it.

  Learning would require adjusting parameters using **training** data and recording the procedure.

Debugging shapes and gradients

> **My remembered idea, example and correction**

Regularisation and early stopping

> **My remembered idea, example and correction**

Batching and epochs

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Input	Hidden representation	Prediction
[4, 2]	[2, 0]	1
[2, 4]	[0, 2]	0
[3, 3]	[0, 0]	0
[5, 3]	[2, 0]	1

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Rectangle inputs finitepositivewidth/heightincm; h=[ReLU(width−height), ReLU(height−width)], weightschosen, nottrained. Predictwider1 iffh1>0; squaretarget 0. Inputs[4, 2]h[2, 0]area 8; [2, 4]h[0, 2]area 8; [3, 3]h[0, 0]area 9; [5, 3]h[2, 0]area 15. Samehwithdifferentareasprevents exact deterministic area prediction forboth. Supported[6, 4]/[7, 5]h[2, 0], wider 1, areas 24/35.

  Sum-only[4, 2]/[2, 4]both 6 loseswiderlabel. Combininghwithsrecoversd=h1−h2, width(s+d)/2, height(s−d)/2; h[2, 0], s8 gives5, 3, area 15. Picturebrightness[5, 2]/[2, 5]orderedh[3, 0]/[0, 3], sum 3 collides. Algebraicfeaturesarenot learned-evidence claims.

- Encode [8, 3], [3, 8] and [4, 4] with the lesson layer. Apply its **prediction** rule and explain the square boundary.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Rectangle inputs finitepositivewidth/heightincm; h=[ReLU(width−height), ReLU(height−width)], weightschosen, nottrained. Predictwider1 iffh1>0; squaretarget 0. Inputs[4, 2]h[2, 0]area 8; [2, 4]h[0, 2]area 8; [3, 3]h[0, 0]area 9; [5, 3]h[2, 0]area 15. Samehwithdifferentareasprevents exact deterministic area prediction forboth. Supported[6, 4]/[7, 5]h[2, 0], wider 1, areas 24/35.

  Sum-only[4, 2]/[2, 4]both 6 loseswiderlabel. Combininghwithsrecoversd=h1−h2, width(s+d)/2, height(s−d)/2; h[2, 0], s8 gives5, 3, area 15. Picturebrightness[5, 2]/[2, 5]orderedh[3, 0]/[0, 3], sum 3 collides. Algebraicfeaturesarenot learned-evidence claims.

- Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Rectangle inputs finitepositivewidth/heightincm; h=[ReLU(width−height), ReLU(height−width)], weightschosen, nottrained. Predictwider1 iffh1>0; squaretarget 0. Inputs[4, 2]h[2, 0]area 8; [2, 4]h[0, 2]area 8; [3, 3]h[0, 0]area 9; [5, 3]h[2, 0]area 15. Samehwithdifferentareasprevents exact deterministic area prediction forboth. Supported[6, 4]/[7, 5]h[2, 0], wider 1, areas 24/35.

  Sum-only[4, 2]/[2, 4]both 6 loseswiderlabel. Combininghwithsrecoversd=h1−h2, width(s+d)/2, height(s−d)/2; h[2, 0], s8 gives5, 3, area 15. Picturebrightness[5, 2]/[2, 5]orderedh[3, 0]/[0, 3], sum 3 collides. Algebraicfeaturesarenot learned-evidence claims.

- The alternative **representation** is s = width + height. Use [4, 2] and [2, 4] to decide whether s alone always supports the wider-than-height task.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Rectangle inputs finitepositivewidth/heightincm; h=[ReLU(width−height), ReLU(height−width)], weightschosen, nottrained. Predictwider1 iffh1>0; squaretarget 0. Inputs[4, 2]h[2, 0]area 8; [2, 4]h[0, 2]area 8; [3, 3]h[0, 0]area 9; [5, 3]h[2, 0]area 15. Samehwithdifferentareasprevents exact deterministic area prediction forboth. Supported[6, 4]/[7, 5]h[2, 0], wider 1, areas 24/35.

  Sum-only[4, 2]/[2, 4]both 6 loseswiderlabel. Combininghwithsrecoversd=h1−h2, width(s+d)/2, height(s−d)/2; h[2, 0], s8 gives5, 3, area 15. Picturebrightness[5, 2]/[2, 5]orderedh[3, 0]/[0, 3], sum 3 collides. Algebraicfeaturesarenot learned-evidence claims.

- For h = [2, 0] and s = 8, recover width, height and area. Does this algebra prove that a **training** program learned the **representation** or that it will work well on noisy measurements?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A different layer uses g = [ReLU(width - height - 1), ReLU(height - width - 1)]. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height.

  Can any deterministic head receiving only g classify both correctly?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Audit a new representation independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled learned **representation**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace an **input** through a hidden **representation** and a decision rule.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Prove an information loss for Learned **representation**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Learned **representation**s. Proposed explanation: "Two hidden values preserve both **input** measurements". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A different layer uses g = [ReLU(width - height - 1), ReLU(height - width - 1)]. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height.

  Can any deterministic head receiving only g classify both correctly?

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- See when a **representation** preserves a distinction and when it destroys it.
- Toy **input** [left, right]; **representation** h=[ReLU(left-right), ReLU(right-left)]. A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/19

A representation can preserve direction

	left	right	h1	h2
A	5	2	3	0
B	2	5	0	3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/19

Learned representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Encode A

A becomes [3, 0], with left brighter.



2. Encode B

B becomes [0, 3], with right brighter.



3. Compare collapsed summary

If the head receives only h1+h2, both become 3 and the target distinction is lost.

CHANGE ONE THING

The head keeps both ordered hidden values instead of their sum.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- See when a **representation** preserves a distinction and when it destroys it.
- Which labelled step matches this detail? If the head receives only h1+h2, both become 3 and the target distinction is lost.
- 1. Encode B
- 2. Encode A
- 3. Compare collapsed summary
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare collapsed summary If the head receives only h1+h2, both become 3 and the target distinction is lost.

A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

Changed-case question and optional hint

- Changed-case practice: The head keeps both ordered hidden values instead of their sum. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. It can distinguish [3, 0] from [0, 3] in this example.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: It can distinguish [3, 0] from [0, 3] in this example. The **representation** retains the direction of brightness difference when coordinates stay separate.

Repair and retain the record

Review Debugging shapes and gradients (ai-5/18). Goal: Check dimension contracts before computing a layer. Short practice: What should a debugging record pair with every gradient value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from **prediction** error. Short practice: Why should the penalty value not be reported as validation error?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Learned representations: the case

AI ACADEMY / AI-5/19 · 01 · LOOK AT THE FACTS

Learned representations: the case

The situation

See when a representation preserves a distinction and when it destroys it.

Supplied evidence

Toy input [left, right]; representation h=[ReLU(left-right), ReLU(right-left)]. A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain learned representations using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

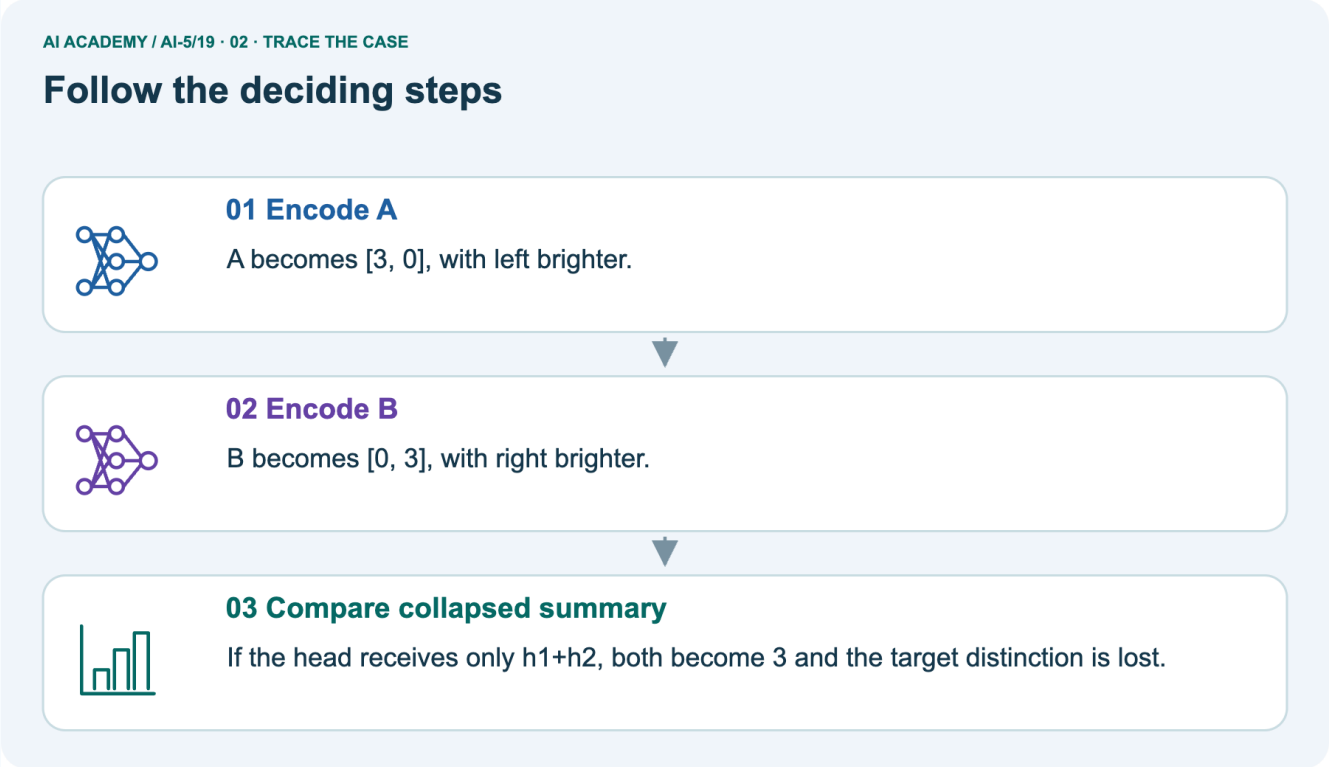
See when a **representation** preserves a distinction and when it destroys it. Supplied facts: Toy **input** [left, right]; **representation** h=[ReLU(left-right), ReLU(right-left)].

A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



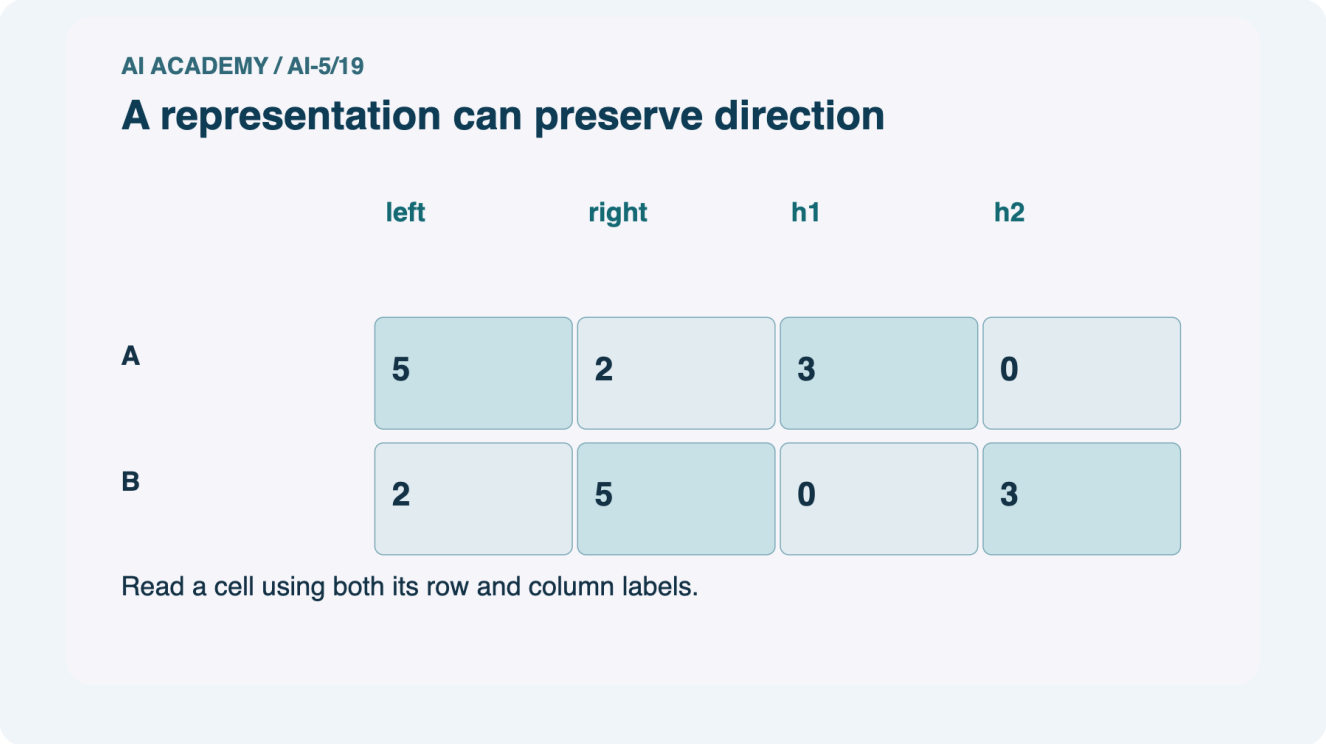
Follow the labelled picture with the matching explanation. Open the vector diagram

Encode A: A becomes [3, 0], with left brighter. → Encode B: B becomes [0, 3], with right brighter. → Compare collapsed summary: If the head receives only h1+h2, both become 3 and the target distinction is lost.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A representation can preserve direction



Follow the labelled picture with the matching explanation. Open the vector diagram

A **representation** can preserve direction. A: left = 5, right = 2, h1 = 3, h2 = 0. B: left = 2, right = 5, h1 = 0, h2 = 3.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / AI-5/19 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

If the head receives only h1+h2, both become 3 and the target distinction is lost.

Changed condition

The head keeps both ordered hidden values instead of their sum.

Worked changed-case result

It can distinguish [3, 0] from [0, 3] in this example.
Why: The representation retains the direction of brightness difference when coordinates stay separate.

Follow the labelled picture with the matching explanation. Open the vector diagram

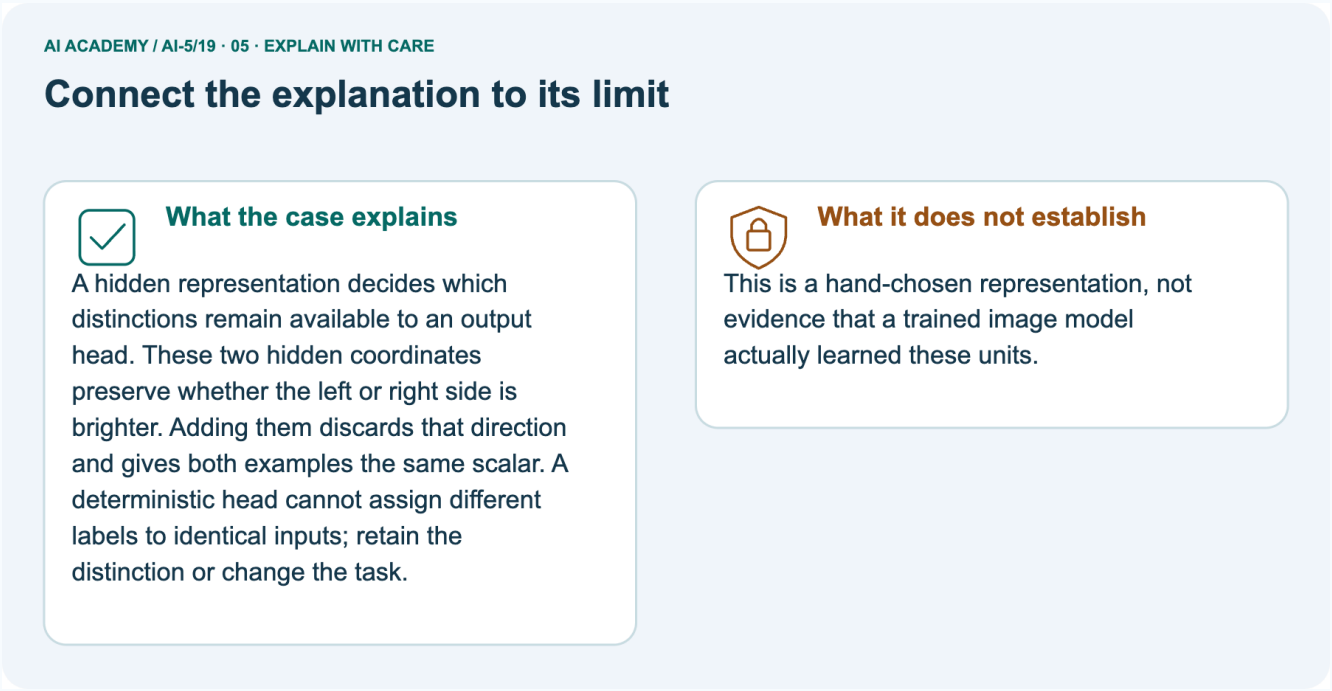
Original case: If the head receives only h1+h2, both become 3 and the target distinction is lost. Changed condition: The head keeps both ordered hidden values instead of their sum.

Expected result: It can distinguish [3, 0] from [0, 3] in this example. Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

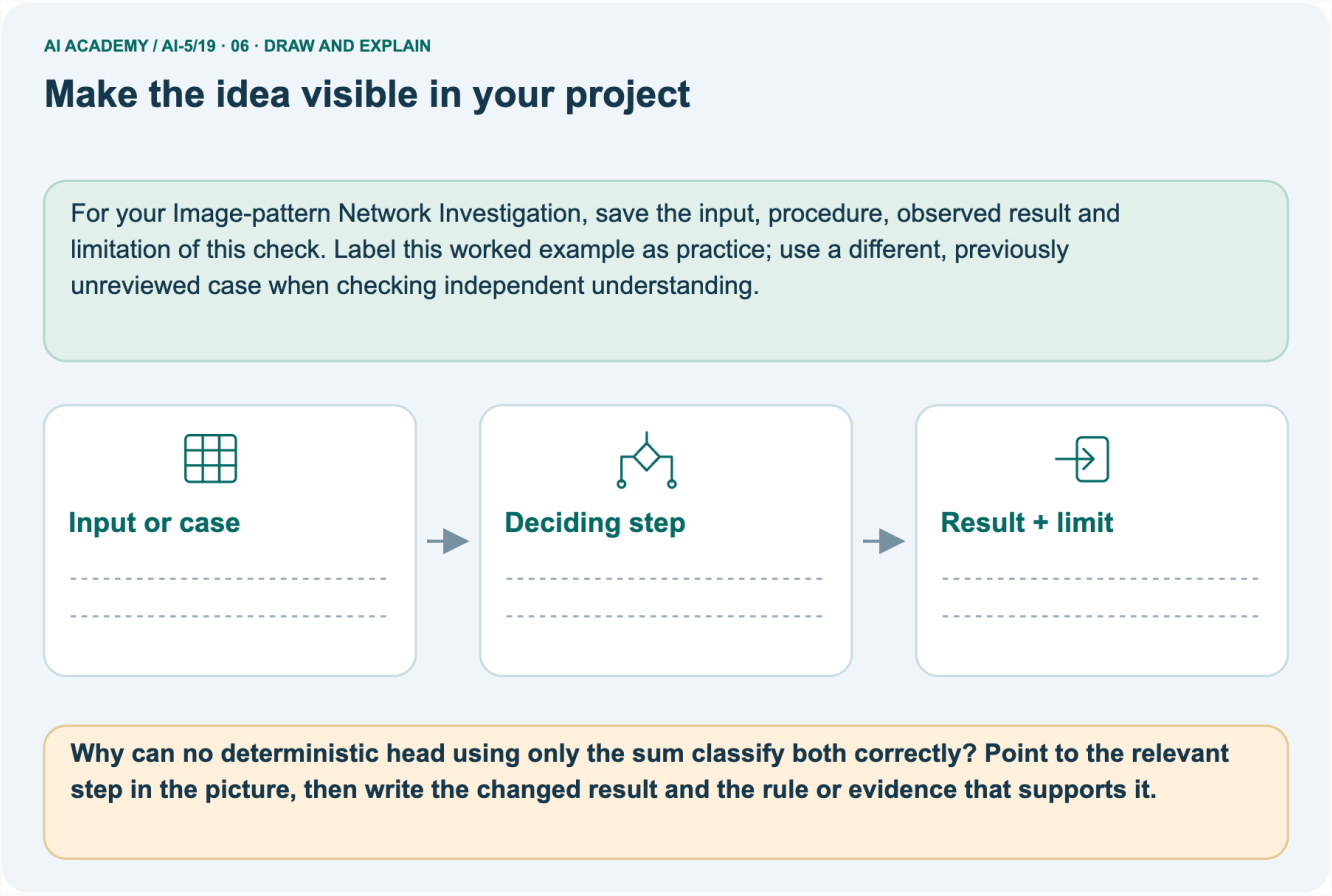
Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

Limit: This is a hand-chosen **representation**, not **evidence** that a trained image model actually learned these units.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Learned representations: the case

02. Follow the deciding steps

03. A representation can preserve direction

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.